41) Give the principle involved in the estimation of halogen in an organic compound by carius method.

42) Give a brief description of the principles of

i) Fractional distillation

ii) Column Chromatography

43) Explain paper chromatography

44) Explain varions types of constitutional isomerism (structural isomerism) in organic compounds

45) Describe optical isomerism with suitable example.

46) Briefly explain geometrical isomerism in alkene by considering 2- butene as an example.

47) 0.30 g of a substance gives 0.88 g of carbon dioxide and 0.54 g of water calculate the percentage of carbon and hydrogen in it.

48) The ammonia evolved form 0.20 g of an organic compound by kjeldahl method neutralised 15ml of N/20 sulphare acid solution. Calculate the percentage of Nitrogen.

49) 0.32 g of an organic compound, after heating with fuming nitric acid and barium nitrate crystals is a sealed tube game 0.466 g of barium sulphate. Determine the percentage of sulphur in the compound.

50) 0.24g of an organic compound gave 0.287 g of silver chloride in the carius method. Calculate the percentage of chlorine in the compound.

51) In the estimation of nitrogen present in an organic compound by Dumas method 0.35 g yielded 20.7 mL of nitrogen at 150 C and 760 mm pressure. Calculate the percentage of nitrogen in the compound

11th Chemistry Chapter-11 Vol-2 EM.indd 158 09-01-2023 14:43:17